Supplies

Electronic components:
Motor
Speed controller
Battery
Receiver and transmitter
Servo

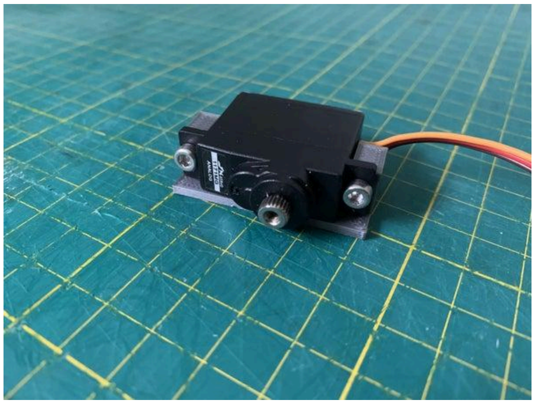
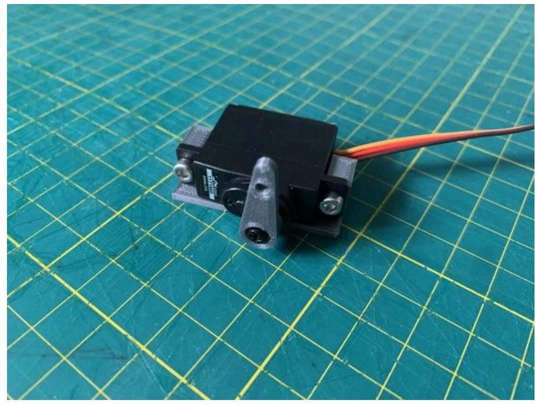
Screws:
All the screws are M3 screws *unless* specified otherwise
4x 8mm
19x 10mm
12x 16mm
8x 22mm
2x 25mm
4x 30mm
4x 40mm
4x nylon locking nut
8x M2.5 5mm set screw
8x M2.5 nut
Other:
3mm Steel Rod (At least 50mm, 25mm each)
9x 623ZZ bearings

Original Instructables File: https://www.instructables.com/3D-Printed-RC-Car/

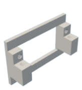
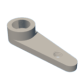
Step 1: Print Chassis

Send the revised chassis to print.

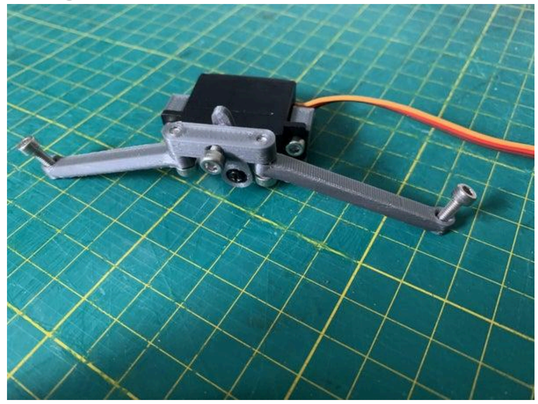
Step 2: Servo

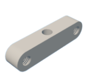
SKIP IF NOT ATTENDING WEEK 2
Attach the servo to the servo mount using two 10mm screws.
Attach the servo arm to the servo with the really small screw. These parts come with the servo motor.

 **servo mount.stl**
 **servo arm.stl**

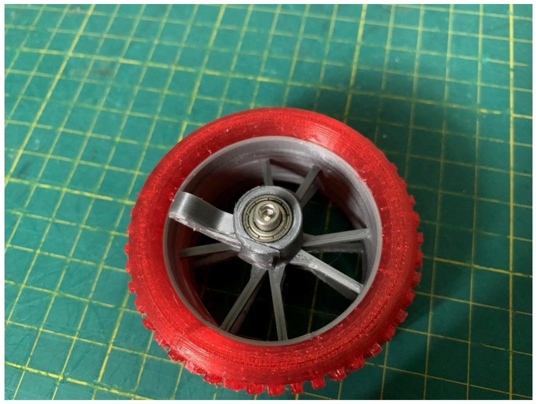
Step 3: Steering Arms

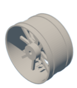
- Screw the Steering Arms to the Center Arm using 10mm M3 screws. The screws come up from the bottom and the arms “point” downward.
- Make sure that the arms can rotate around the screws freely on both ends of both steering arms. If they do not, loosen the screws slightly. If that doesn’t work, file out the inside of the Steering Arm (not the Center Arm).
- Attach the centre part to the servo arm using the pointy screw that came with the Servo.

 **Steering arm.stl**
 **center arm.stl**

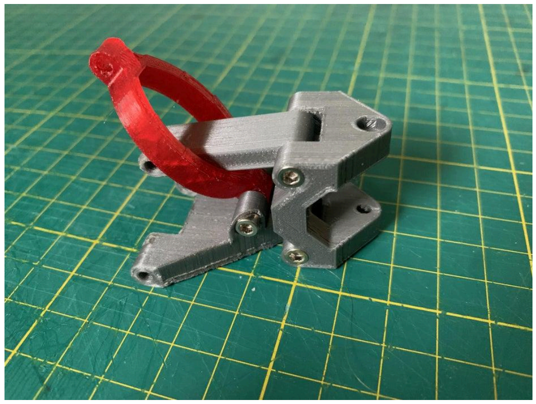
Step 4: Front Wheels



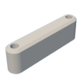
- Press the bearings into the holes (two on each Wheel Bracket). You will probably need to use a vise.
- Put a 30mm screw through the center of the Wheel Bracket, then a washer, then the Rim.
- Screw on a “Nyloc” nut to stop the Rim from coming off.

 **Wheel Bracket.stl**
 **Rim.stl**
 **Tyre.stl**

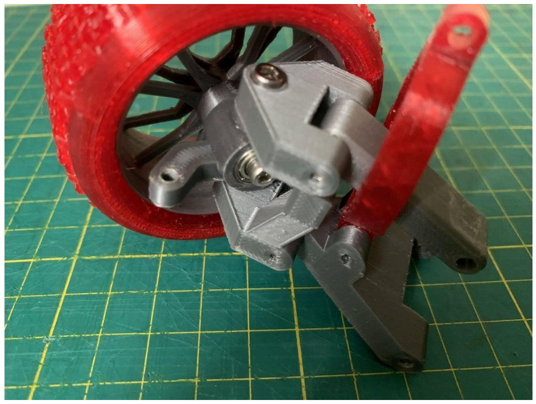
Step 5: Front Arm



- Screw the wishbone (front arm) to the chassis using a 40mm screw.
- Screw the wishbone to the steering bracket using 22mm screws.
- Screw the Front Upper Arm to the steering bracket using 22mm screws.
- Screw the suspension piece to the steering bracket (with the thickest piece towards the wheel) using 18mm screws.
- Make sure that all of these pieces can rotate and move freely.
- PRINT 2 (MIRRORED)

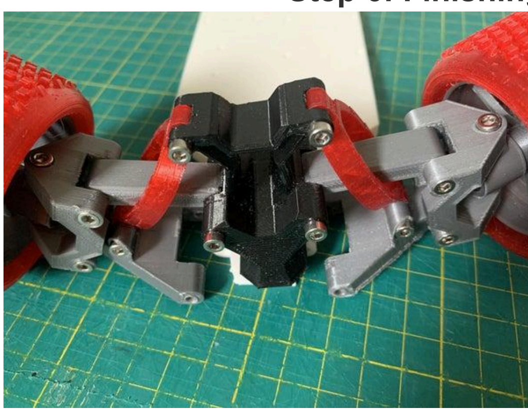
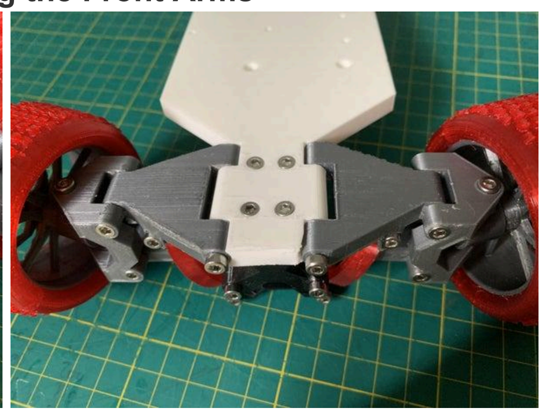
 **Front arm.stl**
 **Front arm top.stl**
 **Steering Bracket.stl**
 **Suspension.stl**

Step 6: Attaching the Wheel



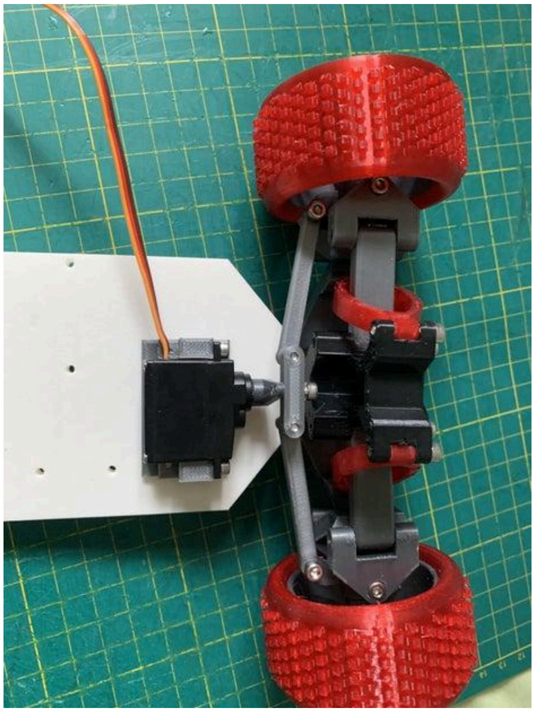
- Screw Steering Bracket to the Wheel Bracket with 8mm screws for the top and bottom.
- Check that the wheel rotates freely, left to right.

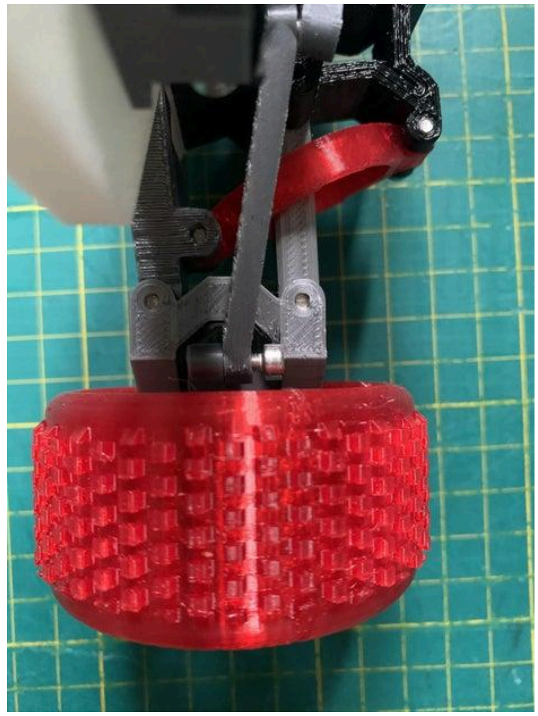
Step 6: Finishing the Front Arms

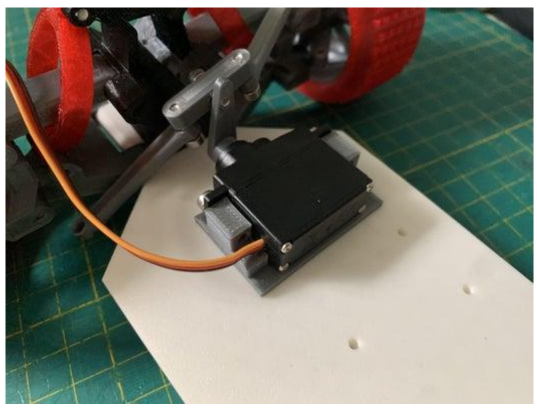



- Screw the suspension to the Front Arm Bracket (this is now part of the chassis print) using 16mm screws
- Screw on the top wishbones using 30mm screws.

Step 8: Finalising the Steering







- Screw the servo to the base using 10mm screws.
- Screw the steering arms to the Wheel Bracket using 10mm screws making sure that there is enough room for the arm to move up and down a little bit.

Step 9: Driver Arm

Print FOUR Driver Arms.

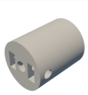
With those Driver Arms, make TWO identical drive shafts:

- Cut 12 mm long pieces of a 2.5mm aluminum rod, to make each pin (you need 2 pins for each drive shaft, so 4 pins in total).
- Glue the two halves of the drive shafts together with super glue, and use blue tape to hold the pieces in place while the glue dries.
- Once the glue is dry (about 10 minutes), put a small drop of super glue in the hole, and then push the pins through the holes, so they look like the picture above.

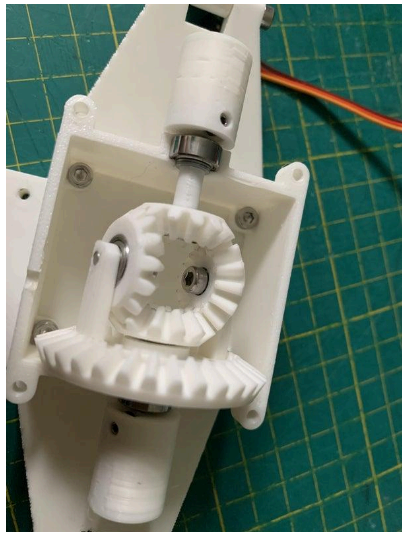
Driver arm.stl

Step 10: Small Gear Assembly

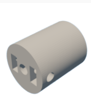
- Cut a 3mm thick aluminum rod to 23mm (use a hand saw with a “metal” blade). A piece of blue tape at 23mm can act as a guide.
- Put the M2.5 nuts in the slots from the top of the Inner Drive Gear, and screw them in with the M2.5 5mm set screw so the threads just barely catch.
- Insert the gear onto one end of the aluminum rod, and tighten the set screws.
- Put one washer onto the aluminum rod, then the Fat Washer, then two washers, and finally a bearing.
- Check for fit by placing the assembly into the gearbox so that when you put on the gearbox connector, it is not too tight (see the image in Step 10).
- Put the M2.5 nuts in the slots of the gearbox connector, and screw then in with the M2.5 5mm set screws so the threads just barely catch.
- Put the gear box connector onto the aluminum rod, and tighten the set screws.

 **Inner Driver Gear.stl**
 **Back gearbox connector.stl**

Step 11: Large Gear Assembly



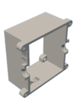
- Assemble Inner Gear Arm Assemblies
  - Put bearings in the Inner Gears (the gears with the hole for the bearing). You may need to use a vise.
  - Screw each Inner Gear to a Gear Arm using 8mm screws making sure that they can spin freely.
- Assemble Large Gear Assembly
  - Cut a 3mm thick aluminum rod to 45 mm (use a hand saw with a “metal” blade or wire cutters in circles).
  - Put in the M2.5 nut in the slots from the top of the Inner Driver Gear, and screw it in with the M2.5 5mm set screw so the threads just barely catch.
  - Insert the Inner Drive Gear onto one end of the aluminum rod, and tighten the set screws.
  - Press a bearing into the Large Gear, and put it onto the aluminum rod, then another washer
- Assemble the Large Gear Assembly
  - Put a drop of super glue into the small rectangular indents on the Large Gear.
  - Press the Gear Arms into the Large Gear, making sure that the gears are lined up, and put it aside so that the glue can dry.

 **Back gearbox connector.stl**
 **Small Gear.stl**
 **Gear Arm.stl**

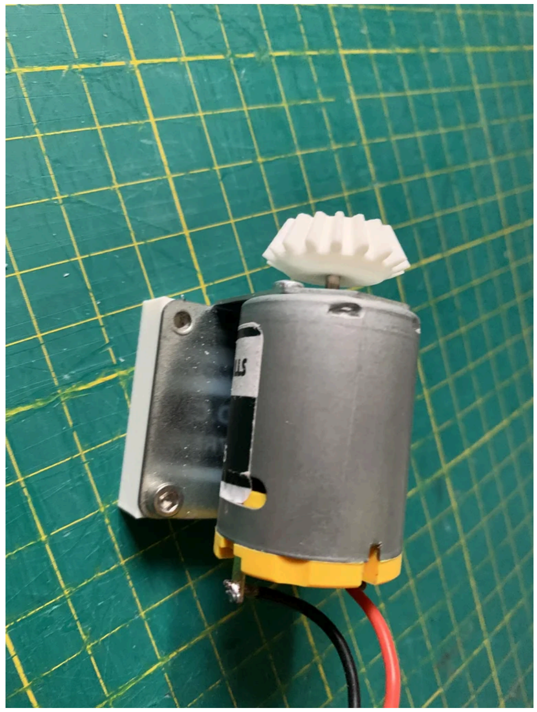
	**Inner Driver Gear.stl**
	**Inner gear.stl**
	**Large gear.stl**

Step 12: Finishing the Gearbox

- Screw the bottom of the gearbox to the Chassis with 10mm screws, and 16mm screws
- Insert all the gears

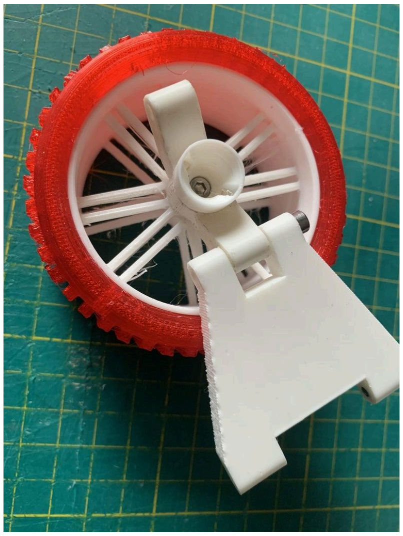
 [Bottom Gear box.stl]

Step 13: Motor



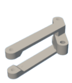
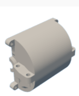
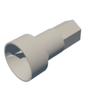
- Screw the motor onto the Motor Mount (not from the Instructable).
- Screw or glue the motor mount onto the chassis, aligning the drive shaft with the small indent in the gear box. This should be as close to the gearbox as possible.
- Put the M2.5 nuts in the slots from the top of the Motor Gear (not in the Instructable), and screw them in with the M2.5 5mm set screw so the threads just barely catch.
- Insert the gear onto the drive shaft of the motor, and tighten the set screws.

Step 14: Back Arm

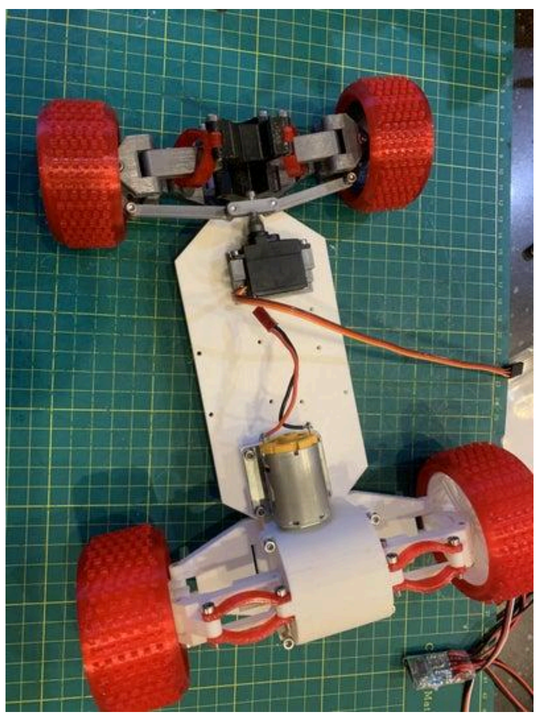
  

- Assemble Rear Wheel Assemblies (2 are needed)
    - Insert the Back Wheel Connector through the Back Wheel Holder.
    - Put an 18mm screw through the Back Wheel Connector.
    - Put the Back Wheel Rim onto the screw, and secure them with a NyLock nuts.
- Assemble Upper Suspension Assembly
    - Screw the Back Top Arms to the top of the Top Gear using 18mm screws.
    - Screw the Back Suspension (there are 4) to the Back Top Arm using 16mm screws.
    - Screw the Back Suspension to the Top Gear Box using 16mm screws.
- Screw the Back Wishbone to the chassis using a 40mm screws.
- Screw the Wheel Holder to the bottom wishbone using a 25mm screw.

Back Tyre Hub.stl

 **Back Suspension.stl**
 **Back top arm.stl**
 **Top gear box.stl**
 **Back wheel connector.stl**

Step 15: Finishing the 3D Printed Parts



- Put the drive shaft in place and screw the top wishbone to the wheel holder using a 16mm screw.

Launchpad’s Tool Glossary

🧹 Post-Processing Tools

1. Files

- Various grits for smoothing surfaces and ensuring parts fit together seamlessly.

2. Flush wire cutter

- For removing supports and cleaning up fine details.

3. Sandpaper

- Useful for cleaning up the schmutz leftover from the layers.
- Wet the sandpaper slightly before use.

🛠️ Assembly Tools

4. Hex Key Set (Allen Wrenches)

- Sizes ranging from 1.5mm to 3mm.
- Use the 2.5mm one.

5. Precision Screwdrivers

- Both Phillips and flathead screwdrivers for handling small screws.

6. Electric Screwdriver

- To expedite the assembly process, especially when dealing with multiple screws.

7. Bench Vise

- For holding parts steady during assembly or when making precise modifications.
- You’ll want this for making sure the bearings are flush into the wheel hub (Step 3).

8. Needle-Nose Pliers

- Useful for gripping small parts, bending wires, or holding nuts in tight spaces.

9. Wire Cutters and Strippers

- Essential for preparing wires when connecting electronic components.

10. Soldering Iron and Accessories

- For connecting wires to motors, batteries, and other electronic components.

11. Multimeter

- To test electrical connections and ensure components are functioning correctly.

12. Lubricants

- Such as grease or oil, to ensure smooth movement of mechanical parts.